TARS: Development of a Drag-Modulating Closed-loop Feedback System to Control a Rocket's Ascent

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This paper presents a study on the development and experimental validation of an advanced air brake system specifically designed for our Targeted Apogee Rocket System (TARS). The primary objective of the air brake is to effectively modulate drag during the coast phase through a closed-loop feedback loop, thereby enabling precise control over the rocket's ascent trajectory. By imposing its influence throughout the flight, the air brake aims to steer the rocket toward a desired apogee. To achieve this, a state-estimation scheme and a model predictive controller are developed, tested, and experimentally validated. The results obtained from the verification and validation process demonstrate the system's effectiveness, solidifying its potential as a solution for achieving ascent control in rocketry applications.

I. Nomenclature

IMU	=	Inertial Measurement Unit
MPC	=	Model Predictive Control
PID	=	Proportional–Integral–Derivative
SRAD	=	Student Researched and Developed
g	=	Gravitational Acceleration
D	=	Drag
m	=	mass
ρ	=	Air Density
T_s	=	Sampling Time
g_x, g_y, g_z	=	Turn Rate Along Axis x, y, z
α	=	Tilt Off Vertical Axis
h	=	Height
v	=	Velocity
δ	=	Flap Deployment Angle
v_∞	=	Far Field Velocity
τ	=	Time Constant
C_{d_f}	=	Coefficient of Drag Of A Flap
C_{d_R}	=	Coefficient of Drag Of The Rocket Body
A_f	=	Reference Area of a Flap
A_R	=	Reference Area of the Rocket Body
Q	=	Process Covariance Matrix
$\mathbf{F}$	=	State Transition Matrix
$\mathcal{P}(h, v, \alpha, \delta)$	=	Apogee Predict Function
DCM	=	Direction Cosine Matrix
$\boldsymbol{q}_k$	=	Quaternion at Time k
Δ_k	=	Error In Reported Height At Time k
R^2	=	Coefficient of Determination

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II. Introduction

The field of rocketry has witnessed remarkable progress since the advent of Robert Goddard's pioneering liquid rocket in 1926. Rockets have now become indispensable in a wide range of military and civil applications. Among these, sounding rockets play a significant role as suborbital vehicles designed to carry scientific payloads and collect data during their flights [1, 2].

In certain scenarios, it is required for sounding rockets to deploy their payloads while still within a substantial atmosphere. To ensure the safety and integrity of both the payload and deployment mechanisms, the rocket's velocity must be minimized during deployment. High velocities can subject the payload to detrimental aerodynamic forces. Consequently, utilizing an *air brake* system on a sounding rocket can help the rocket achieve the desired apogee. An air brake can mitigate the effects of disturbances, enhancing the system's robustness against uncertainties and unmodelled effects.

One notable event within the realm of student rocketry is the Spaceport America Cup. This international competition challenges student teams to design and construct high-powered rockets capable of reaching altitudes of either 10,000 ft or 30,000 ft. The performance evaluation of the rockets heavily weighs the apogee error (difference between the *target* apogee and *actual* apogee). To this effect, many student-built rockets incorporate an air brake as a common addition. However, due to the intricacies involved in developing a comprehensive air brake system, teams often prioritize mechanical aspects, leaving gaps in areas such as state estimation, control development, and system analysis [3–7].

This paper addresses these crucial gaps by delving into the mathematical foundations and practical application of a feedback control system for an air brake in a sounding rocket. Doing so seeks to fulfill the abovementioned needs and offer a more comprehensive understanding of the subject matter. The ***novelties*** of our work are:

- Formulation and validation of a *Kalman filter-based state estimation* scheme specifically for an air brake system. This scheme offers a reliable and accurate method for estimating the system's state, thereby enhancing the overall performance of the closed-loop control system.
- Development and validation of a novel *barometric error correction model* that mitigates undesirable aerodynamic effects caused by a flap deployment. This model enhances the system's accuracy and reliability by minimizing the impact of external disturbances.
- Design of a *Model Predictive Controller (MPC)* tailored for an air brake system. This controller represents an advancement in control techniques for student-built rockets and provides improved system performance, stability, and responsiveness.
- *Successful deployments* and characterization of a *fully integrated air brake system* on a sounding rocket.

The remainder of this paper is organized as follows: Section III briefly describes our rocket's specifications and outlines the air brake system's electrical/mechanical design. We present our state estimator and controller design in Sections IV and V, respectively. Section VI discusses parameter estimation, and Section VII reports results from our in-flight tests. We state our conclusions and future work in Sections VIII and IX.

III. Background

The University of Maryland's student rocket competition team, The Terrapin Rocket Team, designed and built Karkinos for the 2023 Spaceport America Cup. Karkinos, shown in Fig. 1, stands about 12 feet tall and 6 inches in diameter. An 11,500 N-s solid rocket motor powers the rocket and has a nominal predicted apogee of 11,000 ft. The air brake was developed in parallel with Karkinos and was tested along with the rest of the subsystems on March 5th, 2023, and April 2nd, 2023.

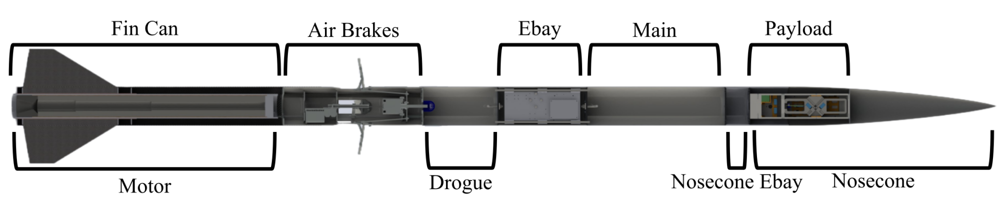

Fig. 1 Karkinos mechanical design layout

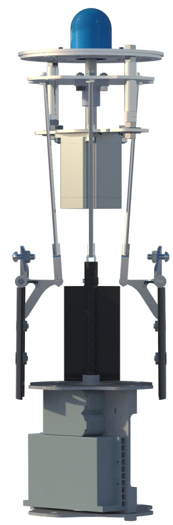

Fig. 2 Flaps fully retracted

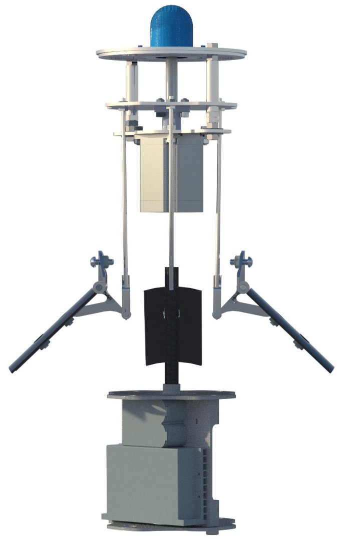

Fig. 3 Flaps deployed to $45°$

Fig. 4 Flaps fully deployed

A. Flap Deployment Mechanism

A stepper motor achieves the deployment of the drag-inducing flaps. The motor displaces an actuator disk, which is connected to four threaded rods. The actuator disk has two channels that restrict the disk to only move in the vertical direction. The rods are connected on their other end to a control horn connected to the flaps. As the disk is displaced down, it rotates the flaps into the airflow. The entire assembly is coupled so that the flaps are mechanically locked to each other and are therefore guaranteed to deploy to the same angle; figs. 2 to 4 shows various deployment configurations.

B. Avionics

The avionics system consists mainly of commercial-off-the-shelf breakout boards and microcontrollers, which play essential roles in the system's overall functioning. The following components contribute to the avionics package:

- LSM9DS1 (IMU)
- BMP388 (Barometric Pressure Sensor)
- Two Teensy 4.1s, one to fuse measurements and compute the control signal and one to command the actuator
- STP-LE23-3H06ADJ (Stepper Motor)
- AMT102-V (Motor Encoder)

IV. State Estimation

An important aspect of a closed-loop control system is obtaining *state-feedback*. In our case, the state $\boldsymbol{x} = [h, v]^T$, where h is the rocket's height and v is its velocity along its longitudinal axis. A system diagram is shown in Fig. 5. To this effect, we use a Kalman filter (KF) as our state estimator. A Kalman Filter serves dual purpose — estimating state feedback and filtering sensor noise.

A. Kalman Filter-based State Observer

The Kalman filter comprises of two steps: *predict* and *update*. It uses the system's process model and prior state to *predict*, and sensor measurements to *update* the prediction, obtaining a "filtered" estimate of the state. The filter is a linear, discrete-time, optimal estimator that assumes uni-modal Gaussian noise. In practice, with proper parameter-tuning, Kalman filters are sufficient for most systems with micro-electro-mechanical sensors (MEMS).

Sensor data from the accelerometer (a_m) and barometer (h_m) are used to drive the process model. The tilt of the

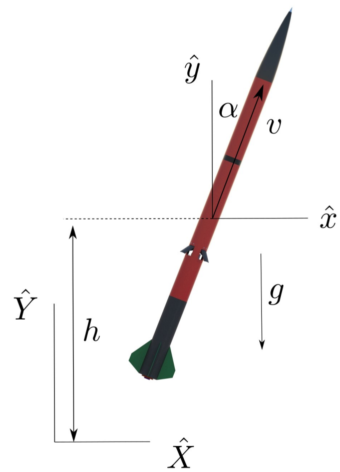

Fig. 5 The relevant states: height h, velocity along body-axis, v and tilt off the vertical-axis, α of the closed-loop altitude control system and inertial axis

rocket, α is estimated in a separate process, as described in the subsequent section. The Kalman Filter assumes the tilt to be already estimated and uses it as one of the inputs. The state variables are as illustrated in Fig. 5, and the corresponding state equations in discrete time are as follows:

Process model:

$$\begin{gathered}\begin{bmatrix} h \\ v \end{bmatrix}_{k+1} = \begin{bmatrix} 1 & T_s \cos\alpha_k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} h \\ v \end{bmatrix}_k + \begin{bmatrix} \frac{T_s^2}{2}(a_m \cos\alpha_k - g) \\ T_s(a_m - g\cos\alpha_k) \end{bmatrix} + \boldsymbol{\omega}_k \\ \boldsymbol{x}_{k+1} = \boldsymbol{F}_k \boldsymbol{x}_k + \boldsymbol{u}_k + \boldsymbol{\omega}_k \end{gathered} \tag{1}$$

Measurement model:

$$\begin{gathered}\begin{bmatrix} h_m \end{bmatrix}_k = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} h \\ v \end{bmatrix}_k + \boldsymbol{\nu}_k \\ \boldsymbol{z}_k = \boldsymbol{H}_k \boldsymbol{x}_k + \boldsymbol{\nu}_k \end{gathered} \tag{2}$$

where T_s is the sampling time, a_m is the component of the acceleration along the axis of symmetry of the rocket, measured from the accelerometer, g is the acceleration due to gravity, and k indicates the time-instance. ω and ν are normal-distributed Gaussian noise with covariance matrices Q and R respectively. R is scalar and Q is as defined in [8],

$$Q = q^2 \begin{bmatrix} \frac{T_s^2}{2} & \frac{T_s^3}{3} \\ \frac{T_s^3}{3} & \frac{T_s^4}{4} \end{bmatrix} \tag{3}$$

where q is the random noise in the accelerometer. In practice, both R and q are tuned with flight data. The actual Kalman Filter algorithm can be found in many common engineering textbooks [9, 10].

B. Tilt Estimation

Although the rocket's tilt (off the vertical) should ideally be implemented into the aforementioned state estimator, and incorporate all nine measurements from the IMU (three each from gyroscope, accelerometer, and magnetometer), a more simple quaternion-based scheme was tested and shown to work for our specific application.

We describe the orientation of the rocket with respect to the earth-fixed frame using quaternions, $\boldsymbol{q} = [q_1\ q_2\ q_3\ q_4]$. We keep track of the rocket's orientation throughout the flight using the following quaternion dynamics.

$$\dot{\boldsymbol{q}} = \frac{1}{2}\tilde{\boldsymbol{\Omega}}\boldsymbol{q} \tag{4}$$

where $\tilde{\boldsymbol{\Omega}}$ is defined as in Eq. (5). The roll, pitch and yaw rates – (p,q,r) are measured from the gyroscope in the IMU.

$$\tilde{\boldsymbol{\Omega}} = \begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix} \tag{5}$$

The quaternion at each time instance, k is computed by forward integrating the dynamics in Eq. (4) using a first-order Euler method.

$$\boldsymbol{q}_{k+1} = \left(I_4 + \frac{1}{2}\tilde{\boldsymbol{\Omega}}T_s\right)\boldsymbol{q}_k \tag{6}$$

where T_s is the sampling time. Note that we normalize the quaternion at each time instance by dividing by its norm. We initialize the initial quaternion, $\boldsymbol{q}_0 = [1\ 0\ 0\ 0]$, implying that the body frame and earth frame are aligned before lift-off.

To compute tilt, we analyze the direction cosine matrix (DCM) between the earth and body frame [11]. The elements of a DCM represent the scalar inner product of the axes of the body with the inertial axes. So by definition, the third row and third column of the matrix, DCM_{33} represents the scalar product between $\boldsymbol{z}^B$ and $\boldsymbol{z}^I$, which is in fact $\cos(\alpha)$, α being the tilt. DCM_{33} in terms of the quaternions is as follows,

$$\mathrm{DCM}_{33_k} = 2(q_{1_k}^2 + q_{4_k}^2) - 1 \tag{7}$$

Rocket's tilt (off the vertical) at each time instance is computed from the obtained DCM_{33} using the following equation.

$$\alpha_k = \cos^{-1}(\mathrm{DCM}_{33_k}) \tag{8}$$

C. Calibration

The only additional calibration is a *bias-correction* for each measurement. After powering up, we ignore a few hundred measurements while waiting for the sensors to reach a steady state. We then average the following few hundred measurements and use this quantity to correct the bias in the subsequent measurements. Due to simplifying assumptions such as zero-mean Gaussian noise, linear process model, and modeling shortcomings, the estimated states are expected to drift while on the pad. To counteract this effect, the computer resets the state to zero every ten seconds if it does not detect a launch.

Fig 6 shows the estimated state as Karkinos sits on the pad, ready for launch. The estimated height and velocity drifts are deemed low enough for our feedback-loop requirements. The drift in the tilt starts at 0.06 degrees per second and becomes almost 0.3 degrees per second close to motor ignition. This is presumed to be caused by the (small) "random-walk" drift of the gyroscope readings. However, due to the periodic *zeroing-out*, the maximum drift we encounter is 3°. In the future, we plan to explore dynamic calibration to counteract this effect.

D. Verfication And Validation

Using simulated data, the calibration and estimator were first synthesized and developed in MATLAB. Once satisfied with the preliminary results, we took raw sensor data from a few flights on small-scale rockets and post-processed them with our MATLAB implementation. Finally, we ran the estimator onboard during the flight and data-logged the estimated states. Post-launch, we compared these *in-flight* estimates to MATLAB post-processing on the raw data and demonstrated that, within precision bounds, they match.

To validate our estimator design, we compared our estimated state to those of some commercial flight computers that flew onboard the rocket as well, Fig. 7 shows flight data curves from our estimated states compared against those estimated by commercial flight computers.

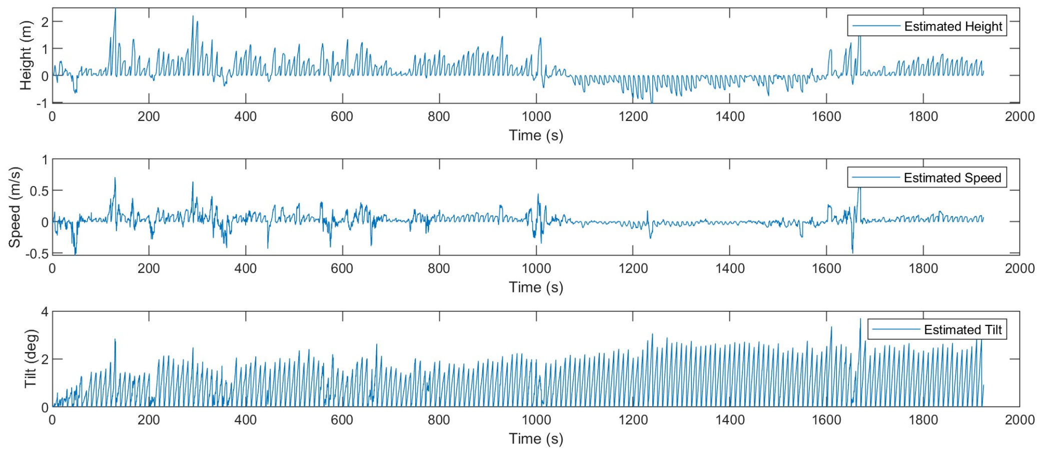

Fig. 6 The estimated state of Karkinos as it sits on the pad. A small drift is apparent in the state estimator. The estimator resets the state to zero every 10 seconds if it does not detect a launch.

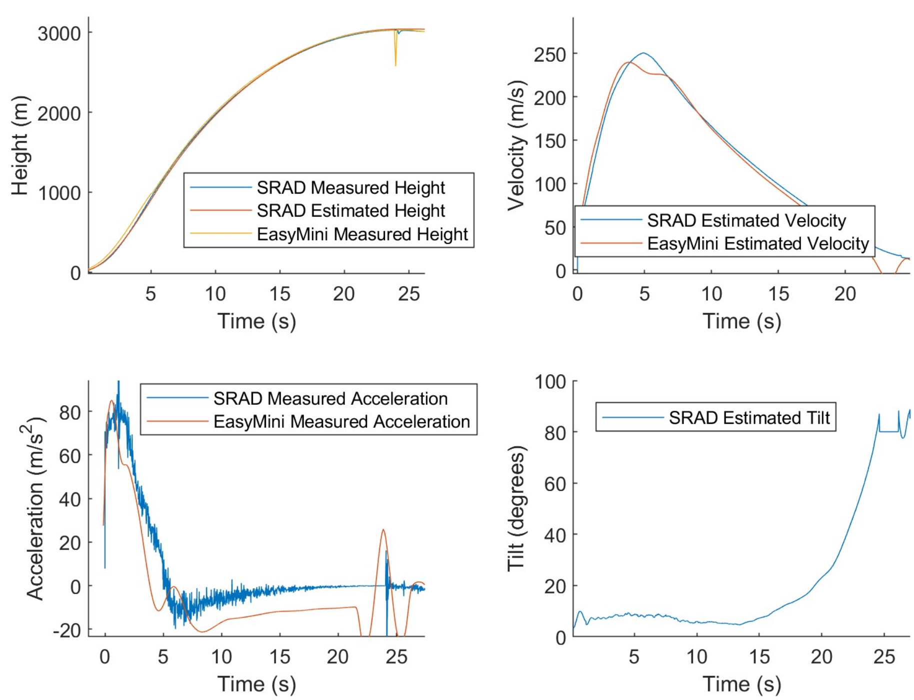

Fig. 7 The estimated states from the flight computer compared to a commercial flight computer (EasyMini) for the Icarus Rocket. (Launch date: 4/2/23)

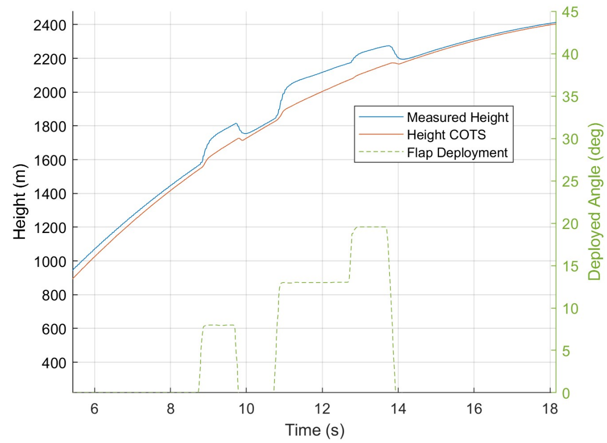

Fig. 8 Barometric sensor reading error while flaps are deployed

E. *Barometric Error Correction*

One of the many challenges to a feedback-control design of air brakes is the fluctuations in barometric pressure around the rocket as the flaps are deployed. Since the altitude measured using the barometer is hugely significant to the Kalman filter, large non-zero-mean noises considerably degrade the estimator's performance. Figure 8 shows the barometric height from the SRAD flight computer and the commercial flight computer that was housed in the electronics bay during a flight with flap deployment. It is evident that when the flaps open, they cause a decrease in pressure around the air brake sensor-suite, and therefore cause an increase in the raw height measurements. The figure also shows a monotonic positive relationship between the flap angle and the “overestimated" height from the barometer.

We also observe this behavior in the commercial flight computer, but to a lesser extent since the commercial flight computer is located farther away from the flaps. This error is especially concerning since this will cause instability in the controller performance. The flaps will initially deploy if it predicts an overshoot. This will cause the measured height to increase, and then the controller will predict an even higher apogee, deploying them farther, thus resulting in a catastrophic positive feedback loop.

Due to the requirement for a rapid, minimum viable solution to this problem, a data-driven heuristic was explored. Though an error correction is usually applied directly to the barometric pressure, we attempt to correct the reported height instead. We model the measured height at any given time as,

$$h_r = h + f(h, v, \delta), \tag{9}$$

where h is the true height, h_r is the height reported by the barometer, and $f(h, v, \delta)$ is the error in height measurement and is a function of true height, velocity, and flap deployment angle. Deployment angle δ, is illustrated in Fig. 9.

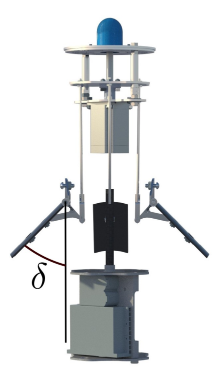

Fig. 9 δ is defined as the flap deployment angle

Figure 10 shows a *wind-tunnel test* with $v_\infty = 50\ m/s$ plotting the reported height from the barometer that was housed inside the air brake module. The average pressure in the wind tunnel did not change significantly during the test but the local pressure near the barometer decreased as the flaps opened. Figure 11 illustrates all of the wind tunnel tests with the reported error as a function of deployment angle.

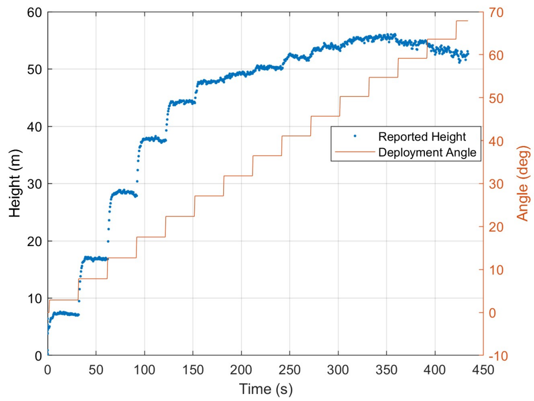

Fig. 10 Barometer height reading while in the wind tunnel with $v_\infty = 50\ m/s$

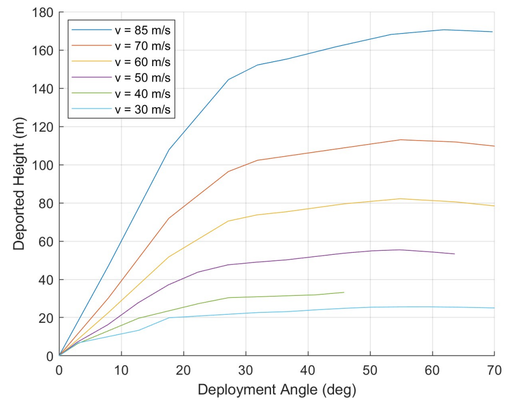

Fig. 11 Height error curves from all wind tunnel tests

Upon careful inspection of the data and trying out a few error models, the following error model gave us the best function fit without overfitting,

$$f(h, v, \delta) = \begin{cases} cq \sin\delta & \delta < 45^\circ \\ cq \sin 45^\circ & \delta \geq 45^\circ \end{cases} \tag{10}$$

where $q = \frac{1}{2}\rho v^2$ is the dynamic pressure and δ is the flap deployment angle. We identify the parameter coefficient, c by fitting this error function to data from wind tunnel tests. Using a standard least squares method, we estimate $c = 0.052$ with an $R^2 = 0.97$. Figure 12 shows the wind tunnel data used for fitting and the surface plot using Eq. 10 with $c = 0.052$.

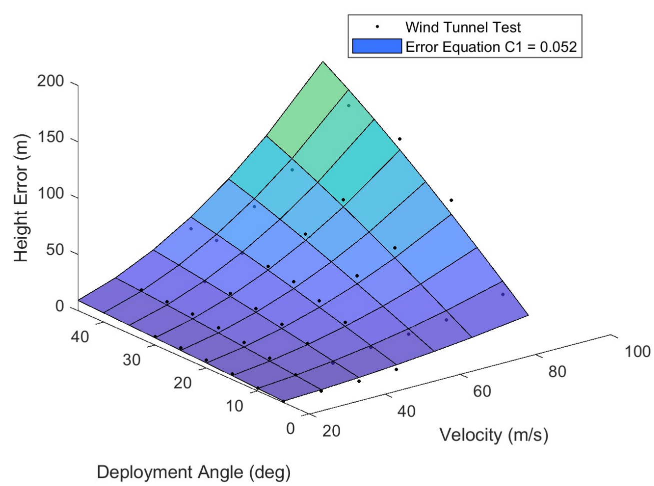

Fig. 12 Surface fit of our error function to the wind tunnel barometer height recordings

Using this error model, we correct the height error before inputting it into the Kalman filter. Figure 13 shows the reported height and the height after the error correction using Eq. 10. It is clear from the plot that although the steady state corrected height is usable, there is a transient time that is significant and Eq. 10 does not capture it. The assumption is that some aerodynamic effects take time to settle. A low pass filter was chosen as a candidate to pseudo-capture these effects. A nominal time-independent low-pass filter will *not* work here since the filter's bandwidth is dependent on the sampling time, and in our case, it is the underlying physics we are trying to capture. With this consideration, we choose a filter defined as,

$$\Delta_{k+1} = (1-\alpha)\Delta_k + \alpha f(h, v, \delta)_k \tag{11}$$

$$\alpha \equiv \frac{T_s}{T_s + \tau} \tag{12}$$

where Δ_k is the value subtracted from the reported height at time k, T_s is the sample time, and τ is the time constant. τ is then tuned using flight data and a value of $\tau = 0.15$ is chosen. Figure 14 shows the flight with the corrected error using this low-pass filter. ***To reiterate the novelty of this method, we can correct for the complicated aerodynamic effects on a commercial barometer with just two tunable parameters: c and τ.***

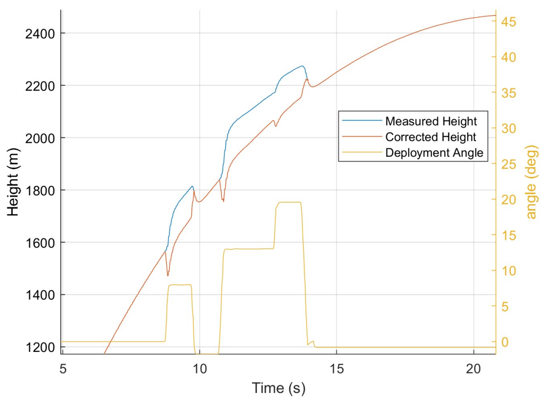

Fig. 13 Height error correction without the low-pass filter

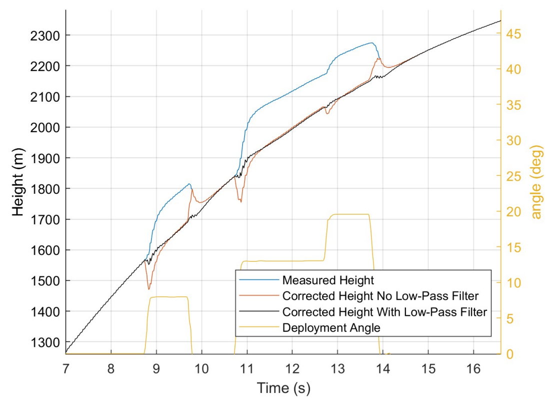

Fig. 14 Height error correction with and without the low-pass filter

This error correction scheme was implemented on Karkinos's full-scale flight test. Figure 15 shows both the estimated states: height and speed. The measured height is highly perturbed due to the changes in pressure around the rocket after flap deployment. The blue solid curve is the estimated height if we did not run any error correction on the reported height and only used our Kalman filter. It is clear that this estimation has a significant error, and our controller would most likely have failed and could have saturated very quickly. Using the low pass filter as defined in Eq. 11 improves the performance of the state estimator. It is worth noting that the nominal Kalman filter can mostly correct and filter noise during the non-transient periods.

V. Controller Design

A few different control schemes were considered and studied, with us ultimately selecting a form of Model Predictive Control **(MPC)** for feedback control. Since the air brake has a limited control time-horizon, a controller that can capture much of the flight dynamics is attractive. Model-free PID controllers were considered but behaved poorly in simulation. Another major benefit of our MPC-based control is its few parameters and no gains to be tuned. The air brake is expensive to full-scale test so getting sufficient flight data sets to properly tune gains is not feasible.

The planning horizon of the MPC is chosen in the velocity space instead of the standard time domain. The MPC plans until velocity goes negative, indicating apogee. We consider a constant control flap-angle (zero-order hold) to simplify the planning optimization. At each controller iteration, the MPC predicts the apogee for various sample flap angles, assuming that the air brake will hold that sampled flap angle for the rest of the flight, and chooses the best flap angle that will get the rocket closest to the desired apogee.

There are three main assumptions about the dynamics of the rocket. First, the angle of attack is always zero; second, the rocket's flight is constrained to a plane; and third, the drag of the rocket body and flaps are decoupled. The second

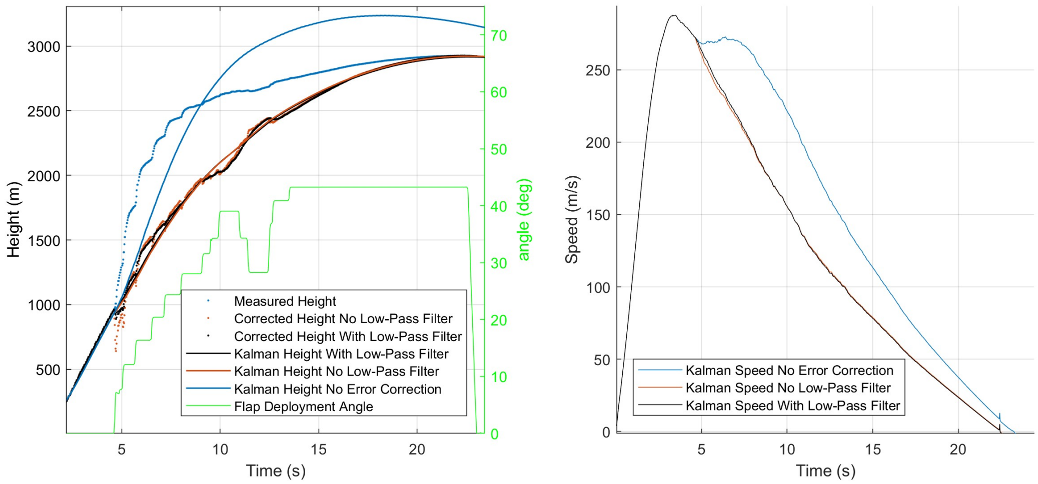

Fig. 15 Estimated states from the Kalman filter with various types of measured height inputs — uncorrected/corrected, with/without low-pass filter

assumption is valid from rocketry experience and the third from wind tunnel tests. With these assumptions, the equation of motion during the rocket's coast phase is,

$$m\ddot{\mathbf{r}} = -\frac{1}{2}\rho(4C_{d_f}A_f(\delta) + C_{d_R}A_R)||\dot{\mathbf{r}}||^2\hat{\mathbf{r}} - mg\hat{\mathbf{y}}, \tag{13}$$

where m is the mass of the rocket, $\mathbf{r}$ is the position vector, ρ is the local air density, C_{d_f} is the coefficient of drag of the flaps, $A_f(\delta)$ is the effective surface area of the flaps and is a function of the flap angle, C_{d_R} is the coefficient of drag of the rocket body, A_R is the reference area of the entire rocket, and g is the acceleration due to gravity. The constant '4' appears in the equation since there are four flaps in our air brake.

Note that $||\dot{\mathbf{r}}||^2 = \dot{x}^2 + \dot{y}^2$. Using the three aforementioned assumptions, and further assuming that ρ is locally constant, we write out the rocket's equations of motion as,

$$\ddot{y} = -\frac{1}{2m}\rho(4C_{d_f}A_f(\delta) + C_{d_R}A_R)\dot{y}\sqrt{\dot{x}^2 + \dot{y}^2} - g, \tag{14}$$

$$\ddot{x} = -\frac{1}{2m}\rho(4C_{d_f}A_f(\delta) + C_{d_R}A_R)\dot{x}\sqrt{\dot{x}^2 + \dot{y}^2}. \tag{15}$$

At any point in the flight, we forward integrate Eq. 14 and 15 with the initial conditions being the current state to obtain apogee predictions. The initial conditions are computed from the state estimates obtained using the Kalman filter,

$$\begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \end{bmatrix}_0 = \begin{bmatrix} 0 \\ v\sin\alpha \\ h \\ v\cos\alpha \end{bmatrix} \tag{16}$$

where h, v, and α are the rocket's estimated height, velocity, and tilt at the current time instance. Using this formulation, we define an apogee predict function $\mathcal{P}$,

$$h_{max} = \mathcal{P}(h, v, \alpha, \delta) \tag{17}$$

The function takes in height, velocity, tilt, and a flap deployment angle and integrates Eq. 14 and 15 until $\dot{y} \leq 0$ (descent-phase). Recall that $\mathcal{P}$ assumes a constant δ. It returns $y(h_{max})$ at the time instance when $\dot{y}$ first becomes negative as the predicted apogee. Equation 17 was implemented with a second-order Runge-Kutta integrator since it gives a sub-meter precision compared to the more standard fourth-order but runs faster on the resource-limited flight computer.

Using $\mathcal{P}(h, v, \alpha, \delta)$, the MPC-based planner determines the *optimal* flap-angle, δ that will result in h_{max} being close to the desired apogee. We use a binary search method with a 5° resolution to search over the space of control inputs (flap-angles). To verify the veracity of the apogee predict function $\mathcal{P}$, we flew the flight computer on a rocket without actuating the flaps and computed the predicted apogee during the coast phase assuming $\delta = 0$. Figure 16 is the flight of Karkinos from March 5th, 2023. It was Karkinos in its full configuration with the air brake flaps turned off. The yellow curve is the output of Eq. 17 during the coast phase. Figure 17 is the error of the predicted apogee function compared to the actual apogee.

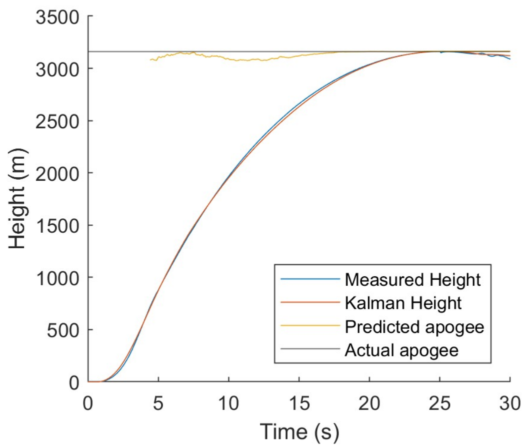

Fig. 16 Estimated height and predicted apogee in time on the Karkinos flight of March 5th

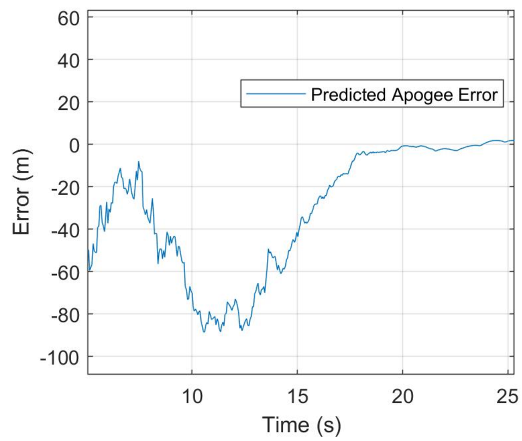

Fig. 17 Error in the predict apogee function of the controller

VI. Parameter Estimation

The apogee predict function $\mathcal{P}$, requires the estimation of constants: m, C_{d_f}, $A_f(\delta)$, C_{d_R} and A_R — the mass and drag characteristics of the rocket and the flaps. m is simply the empty mass of the rocket. It is calculated as the measured wet mass minus the expected propellant mass. Next, consider C_{d_R} and A_R. We can obtain estimates of these values from our OpenRocket model but OpenRocket does not have all of the details of the rocket, such as surface finish and various exposed hardware. Another option is using a CFD tool to estimate C_{d_R}, but this has the same downside as OpenRocket with the additional issue that our usage of the simulation software is not validated. Lastly, we can estimate C_{d_R} using our wind tunnel tests. There are two problems with using the wind tunnel tests: the measured drag includes the model mount, and we were unsure how to subtract it accurately. Second, wind tunnel tests are expensive and logistically complicated. So, it was determined not to be a long-term solution as rocket designs can change much more rapidly than our ability to run wind tunnel tests. Our solution was to estimate these constants *in-flight* using onboard sensors. Consider the acceleration due to drag without flap deployment, written as,

$$a_D = \frac{1}{2m}\rho C_{d_R} A_R v^2. \tag{18}$$

a_D is precisely what the IMU measures while the rocket is in the coast phase. Given the fact that C_{d_R} and A_R only ever appear as a product in our controller equations, we can readily estimate it from Eq. 18 as,

$$C_{d_R} A_R = \frac{2ma_D}{\rho v^2}. \tag{19}$$

In practice, a_D is generally noisy, so we take the mean of $C_{d_R} A_R$ over one second of measurements after motor burnout and before flap deployment. This method was within 15% of the value obtained from OpenRocket.

The drag characteristics of the flaps: C_{d_f} and $A_f(\delta)$, were determined from wind tunnel tests with the assumption,

$$A_f(\delta) = A_0 \sin\delta, \tag{20}$$

	Wind Tunnel	OpenRocket	In flight sensors	CFD
C_{d_R}	0.71	0.51	0.58	0.66
Drawback(s)	Includes mount	Missing rocket characteristics	Noisy measurements	Low fidelity model and simulation tool complexities

Table 1 Coefficient of drag of the rocket assuming a $29.9in^2$ reference area

where $A_0 = 10in^2$ is the approximate surface area of each flap, and δ is flap-deflection angle. Using the standard drag equation and assuming the drag of the rocket and flaps are decoupled, we can write the equation for the total drag as,

$$D = \frac{1}{2}\rho(4C_{d_f}A_0 \sin\delta + C_{d_R}A_R)v^2. \tag{21}$$

This equation was fit using a linear least square method to the data from wind tunnel tests. This method resulted in estimated $C_{d_f} = 0.95$ with a $R^2 = 0.996$. Figure 20 shows the data points as well as the contour fit of Eq. 21 using $C_{d_f} = 0.95$.

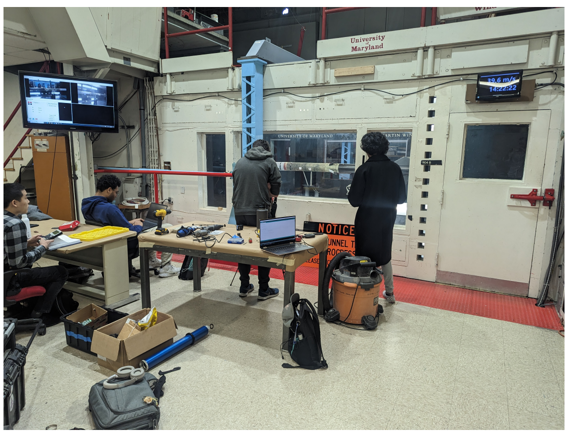

Fig. 18 Observation room in the wind tunnel

Fig. 19 Karkinos mounted in the wind tunnel

VII. In-Flight Tests and Validation

Karkinos had two full integration tests, one on March 5th, 2023, and the second on April 2nd, 2023. In the former flight, we disabled the air brake deployment, enabling it in the latter one. The first flight reached an apogee of 3,164 m or 10,381 ft. We turned on the air brake deployment for the second flight and set the desired apogee to 2,900/m or 9,514/ft. On the second flight, Karkinos achieved an apogee of 2,922 m or 9,588 ft, ***an apogee error of just* 74 *ft!***

We integrated the rocket state from just after motor burnout to validate our process model according to Eq. 13. We integrated considering the flap deployment profile as recorded on the April 2nd flight and another assuming zero flap deployment. We then compared these two curves with the two actual flight profiles. Figure 21 shows these four curves. ***It is clear that the process model is close to the true rocket dynamics and that we shaved off nearly 800 ft to reach our desired apogee!*** We hypothesize that the red curve's divergence at the beginning of the flight is due to our barometric error correction limitations.

VIII. Conclusion

An air brake system does not operate as a nominal feedback-control actuator mechanism; it can solely increase drag and does not possess the capability to add energy to the system. An air brake also has the most control authority during the early stages of the flight when the velocity is large, but the apogee prediction is the most uncertain. Hence, it

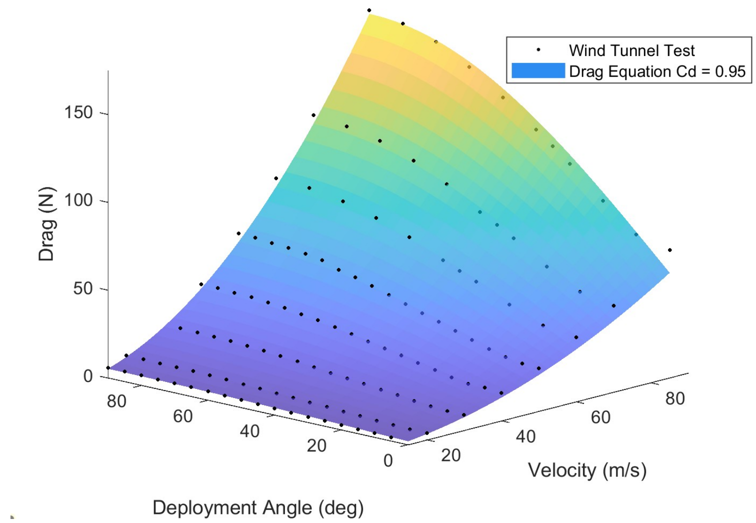

Fig. 20 Wind tunnel test and resultant fit of the drag equation

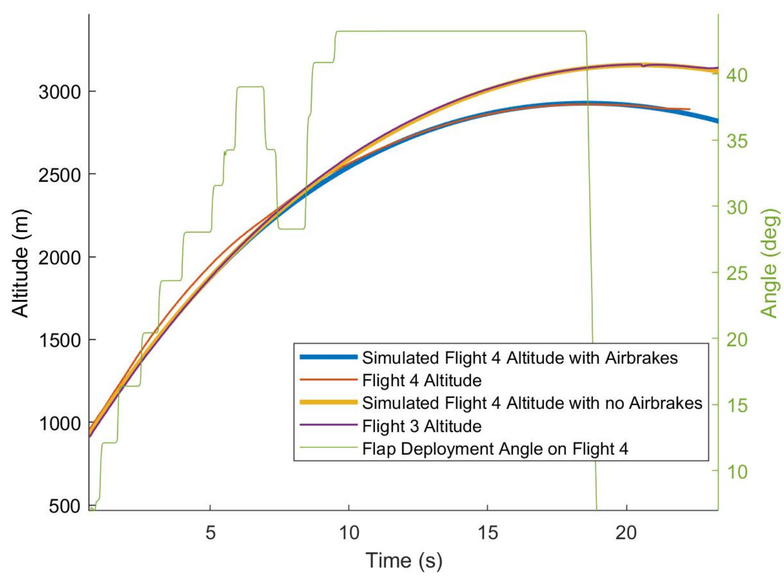

Fig. 21 The blue and orange curves are the integration of the initial condition of the rocket (flight 4) just after motor burnout. The blue considers the drag from the deployed flap angle, and the orange assumes no flap deployment. These curves are compared to actual flights 4 (red) and 3 (purple). Note that flights 3 and 4 were nearly identical except for flight 3, having the air brakes disabled.

becomes crucial to exercise caution during the early stages of flight to prevent over-actuation, which could result in an undershoot.

A model predictive controller is deemed appropriate to address the complexities of the air brake system and accurately predict its impact on the flight profile. This controller considers the full effect of flap deployment, albeit necessitating an accurate estimation of the system's state. Even minor errors in state estimation can lead to significant errors in the model prediction function. Consequently, our approach includes using a barometric error correction model to mitigate height measurement errors caused by the flaps' interaction with the airflow.

Proper estimation of the rocket's mass and drag characteristics, as well as the drag characteristics of the flaps, is essential for the closed-loop system's performance. This parameter estimation is achieved through a combination of wind tunnel tests and onboard estimations, ensuring system behavior accuracy.

Through comprehensive analysis and experimentation, this paper demonstrates the robustness and efficacy of our approach, emphasizing the importance of accurate state estimation and the utilization of an appropriate controller design. The various feedback system techniques outlined in this paper enable us to achieve accurate state estimates and help realize our model-based controller design. Under the validity of assumptions and for our rocket's operating regime, the proposed closed-loop controller successfully equips our rocket to reach a desired apogee; ensuing a **T**argeted **A**pogee **R**ocket **S**ystem (**TARS**).

IX. Future Work

Currently, the state estimation scheme has a few limitations. The authors intend to investigate the use of an integrated Extended Kalman Filter or Unscented Kalman Filter as an alternative state observer to improve the performance of the state estimation. An EKF/UKF also has the benefit of more properly coupling the tilt estimation with the rocket's 3D position and velocity estimates. Additionally, the barometric error correction model must be addressed more formally. A model that directly corrects the barometric pressure will probably capture the aerodynamic effects more closely.

Although the controller design works, it makes some significant assumptions. Improving the apogee-predict function, $\mathcal{P}$, to integrate higher-order terms will potentially increase the controllability of the system; a few extensions are a non-static coefficient of drag and better wind models. Modeling uncertainties can be considered in the controller design to better-bound performance and limit over-actuation.

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